
Calculating Limits Using the Limit Laws

Work these in class. Worked solutions are on the posted notes page — each problem below names the section heading to look for.

2.3 • Problem 1 (worked together)*see: The One Time We Use Them*

Watch the *shape* before you watch the algebra: one move per line, and on the right the law that licenses it — named only so we can point at it.

2.3 • Problem 2*see: Why You Can Usually Just Plug In*

Exam 1 asks this one almost word for word, and it says **justify your steps**. Evaluate $\lim_{x \rightarrow 3} (2x^2 - 5x + 1)$ — and write the justification you would hand in.

2.3 • Problem 3*see: Cancelling Is Division*

Solve $x^2 = 3x$. Cancel the x from both sides and you get $x = 3$. Now check — is that *every* solution?

2.3 • Problem 4*see: These Are Not the Same Function*

Let $f(x) = \frac{x^2 - 9}{x - 3}$ and $g(x) = x + 3$. What is $g(3)$? What is $f(3)$?

2.3 · Problem 5*see: Factor and Cancel*

Finish it: $\lim_{t \rightarrow 4} \frac{t^2 - t - 12}{t - 4}$ — and justify every line, the way Exam 1 will ask.

2.3 · Problem 6*see: Two More Ways to Rewrite*

$\lim_{x \rightarrow 0} \frac{\sqrt{x+9} - 3}{x}$ — same rule: a justification on every line.

2.3 · Problem 7*see: Closing the Trap*

Both outer functions are polynomials. Take their limits as $x \rightarrow 0$, and then say what the Squeeze Theorem lets you conclude.

2.3 · Problem 8*see: Trapped Is Not Enough*

The bounds are $f(x) = -1$ and $h(x) = 1$. Does the Squeeze Theorem apply?